Three studies in rotational dynamics

Gimballed gyroscope, spinning top and torque-free tumbling body

A spinning rotor couples the motion of two gimbals. A top precesses under gravity. A free body flips over and over without any applied torque. Each motion was simulated in ManifoldsX for 12 s at fixed steps of 1, 0.5 and 0.25 ms and compared with conservation laws or an exact solution. All nine runs finished without retries or step cuts.

Table 1. Main results at a 1 ms step

Study	Result
Gimballed gyroscope	Energy and the two conserved momentum components stay constant to 10^{-11} or better. Halving the step changes the orientation by at most 8.4×10^{-11} rad.
Spinning heavy top	The motion follows the exact steady-precession solution to within 3.7×10^{-8} rad.
Tumbling body	The body flips four times in 12 s, and its angular velocity matches the analytical solution to within 1.6×10^{-9} rad/s.

1. Gimballed gyroscope

Léon Foucault gave the gyroscope its name in 1852, when he used a fast rotor in gimbals to show the rotation of the Earth. Gimballed rotors later became the heart of the gyrocompass and of inertial navigation. In our model the rotor sits inside two freely moving gimbals. It starts spinning, and the whole assembly is given an initial rotation about the outer hinge. The spinning rotor then couples the motion of the two gimbals.

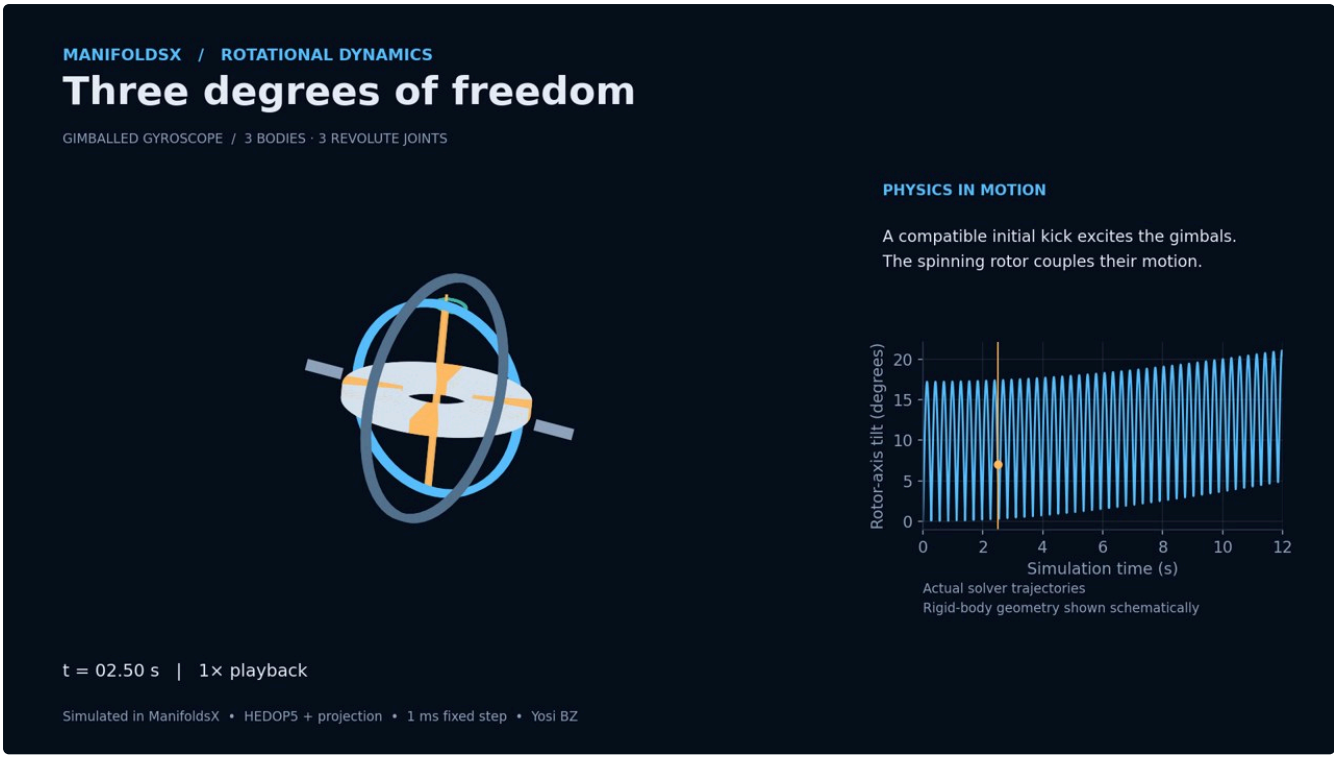


Figure 1. The simulated gyroscope. The shapes are schematic; the masses and inertias are those listed below.

Gimbals	Mass 1 kg each; inertia diag(0.1, 0.1, 0.1) kg m^2.
Rotor	Mass 5 kg; inertia diag(0.3, 0.3, 0.5) kg m^2.
Start	Outer and inner gimbal angular velocity (3, 0, 0) rad/s; rotor (3, 0, 20) rad/s; hinge axes along X, Y and Z.
Setting	All centres of mass at the pivot; no gravity and no bearing friction. Angular momentum about the free X axis, rotor axial spin and total energy are conserved.

Table 2. Largest change over 12 s at 1 ms

Quantity	Change
Mechanical energy (relative)	1.2×10^{-12}
Angular momentum about X (kg m²/s)	2.4×10^{-11}
Rotor axial speed (rad/s)	2.3×10^{-11}

Halving the step changes the orientation of any body by at most 8.4×10^{-11} rad. With the conserved quantities constant to 10^{-11} or better, the motion is converged in step size.

2. Spinning heavy top

Lagrange solved the symmetric heavy top in his *Mécanique analytique* of 1788. Its simplest motion is steady precession, the slow circling every gyroscope demonstration shows. The top spins about its own axis while that axis travels around the vertical. Its initial motion is set for steady precession, so the tilt never changes. That gives an exact solution for both the axis direction and the full orientation of the body.

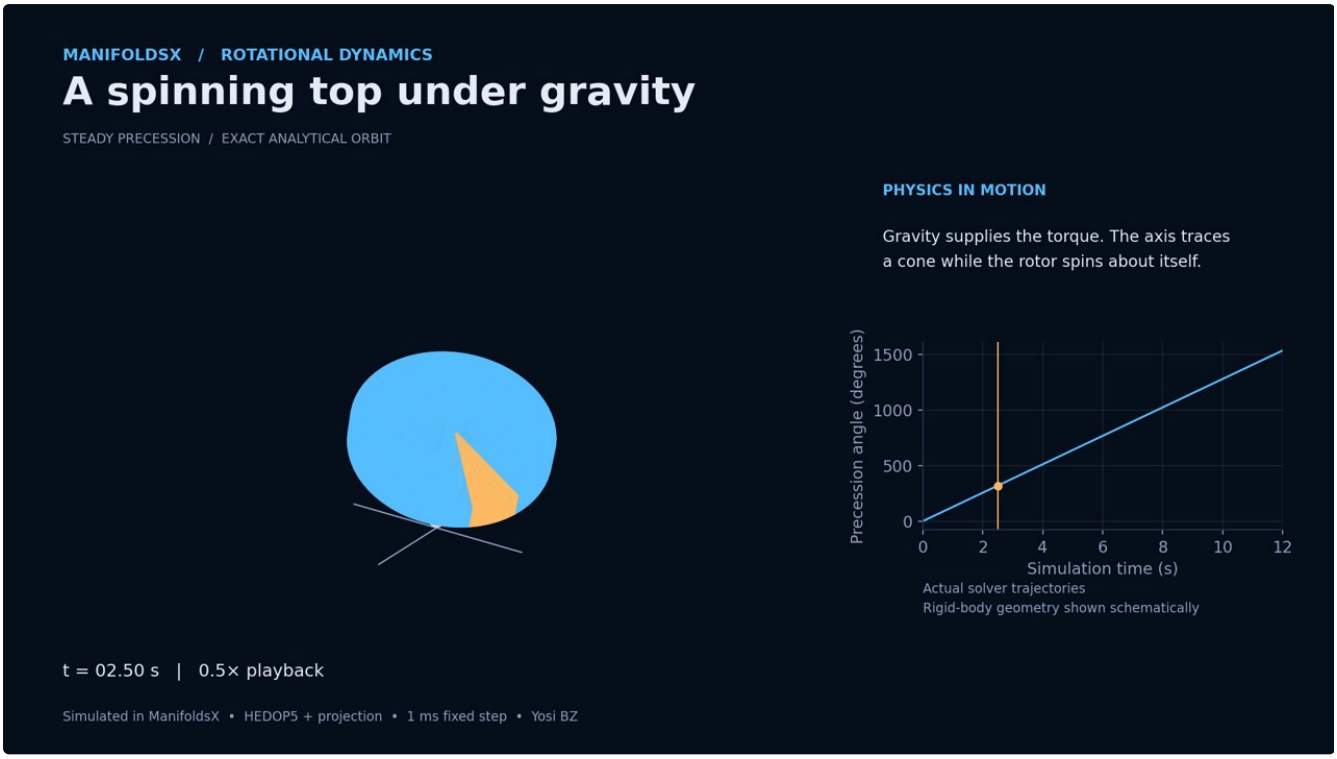


Figure 2. The simulated top. First 6 s, half speed in the animation.

Body	Mass 1 kg; pivot to centre of mass 0.2 m; principal inertias (0.02, 0.02, 0.01) kg m^2.
Start	Tilt 0.45 rad; axial spin 100 rad/s; gravity 9.81 m/s^2.
Exact motion	Precession at 2.23 rad/s about the vertical, with the tilt fixed at 0.45 rad.

Table 3. Largest error against the exact solution, 1 ms step

Quantity	Error
Full orientation (rad)	3.7×10^{-8}
Symmetry-axis direction (rad)	4.3×10^{-9}
Mechanical energy (relative)	1.4×10^{-11}

The orientation error falls from 3.7×10^{-8} rad at 1 ms to 3.8×10^{-11} rad at 0.25 ms, about 31 times per halving of the step, as expected for a fifth-order method.

3. Tumbling body

Euler’s rigid-body equations of 1765 predict that spin about the middle principal axis is unstable. In 1985 cosmonaut Vladimir Dzhanibekov saw it happen aboard the Salyut 7 station, when a wing nut he had spun off its bolt kept flipping over in mid-air. Our body starts spinning almost exactly about its middle principal axis. That rotation is unstable: a small disturbance grows until the body flips, and the flip repeats with no torque acting.

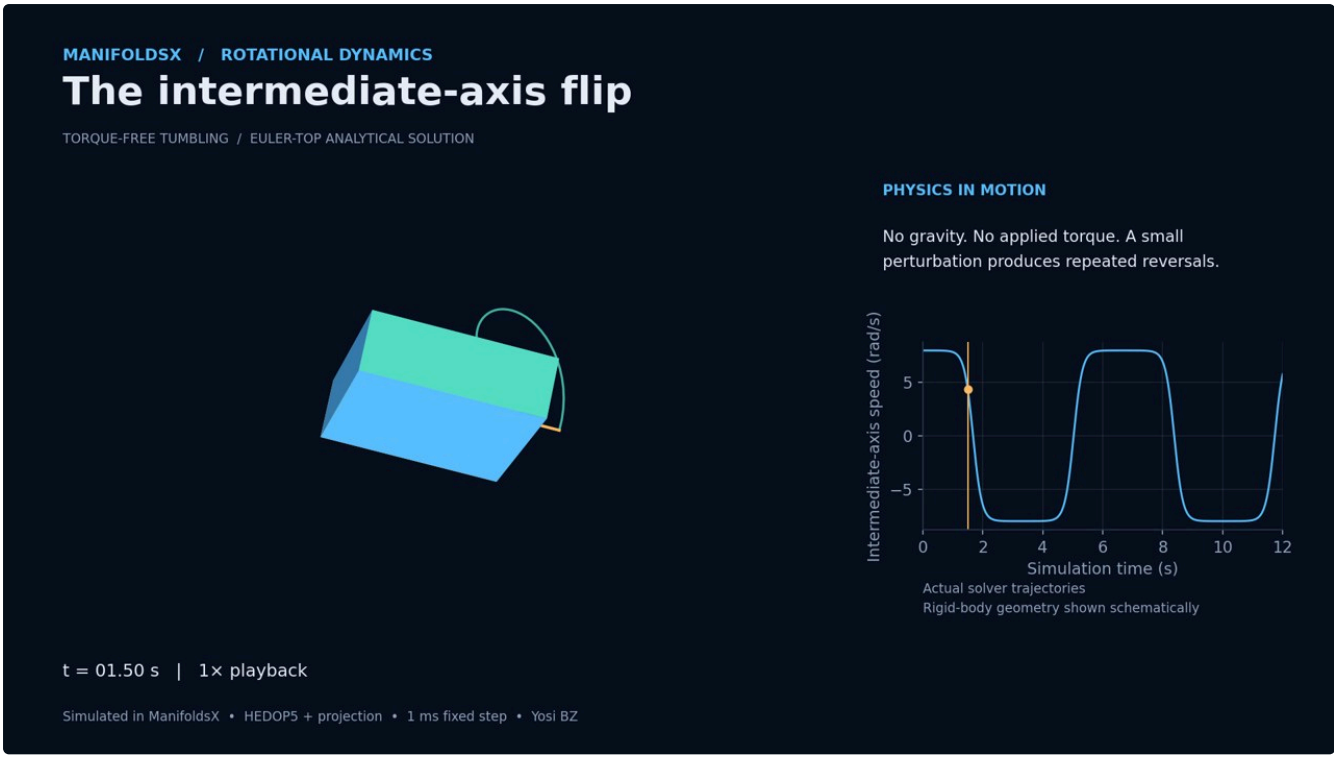


Figure 3. The simulated body. The cuboid has the same principal inertias as the model.

Body	Mass 1 kg; principal inertias (0.2, 0.31, 0.47) kg m^2.
Start	Body angular velocity (0, 8, 0.05) rad/s; no gravity, joints or applied forces.
Reference	The exact Euler solution in Jacobi elliptic functions, evaluated at 45-digit precision and compared every 5 ms.

Table 4. Results over 12 s at 1 ms

Quantity	Value
Flips of the middle-axis spin	4
Angular-velocity error (rad/s)	1.6×10^{-9}
Angular momentum and energy (relative change)	round-off

4. Step refinement

Table 5. Largest error at each step size over 12 s

Step (ms)	Top orientation (rad)	Tumbling angular velocity (rad/s)	Gimbal, change on halving (rad)
1	3.7×10^{-8}	1.6×10^{-9}	8.4×10^{-11}
0.5	1.2×10^{-9}	1.3×10^{-9}	5.2×10^{-12}
0.25	3.8×10^{-11}	4.9×10^{-9}	–

The top error falls quickly with the step. The tumbling body is already within a few nanoradians per second of the exact solution at 1 ms, and smaller steps do not improve it further.

5. Settings

All runs use the HEDOP5 integrator with position and velocity projection, at fixed steps. Projection tolerances are 10^{-13} for position and 10^{-12} for velocity, with up to ten position iterations. The solver works in double precision. Gimbal results use the largest orientation difference across all three bodies; top errors are the rotation between simulated and exact orientation; tumbling errors are the difference of the three angular-velocity components. Model files, reference calculations and raw results are available on request.

References

[1] J. Peraire and S. Widnall, MIT 16.07 Dynamics, Lecture L30: 3D rigid body dynamics, tops and gyroscopes, 2009.

[2] NIST Digital Library of Mathematical Functions, Section 22.13, Jacobi elliptic functions. dlmf.nist.gov/22.13
